

Puzzle Design — Teacher Guide

6th Grade STEM Field Trip Lesson Plan: Puzzle Design Project (Single-Day, 4-5 Hours)

Grade Level: 6th Grade Subject/Area: Integrated STEM (Technology, Engineering, Art, Mathematics) with Arizona Standards Duration: 4-5 hours (one full field trip day, e.g., 9:00 AM-1:00/2:00 PM, including breaks and transitions) Location: Field trip at Ft. Apache Building 106 with access to computers/laptops (Inkscape installed) and a laser cutter/engraver (adult-supervised). If computers are limited at the site, students may bring pre-sketch ideas or complete heavy digital work in rotating small groups. Project Overview: Students will design a custom 2D jigsaw puzzle in Inkscape inspired by cultural and modern geometric art (Navajo rugs, Anasazi/Pueblo pottery, pictographs/symbols, contemporary patterns). They will apply math to scale designs to a precise 15 cm × 20 cm (150 mm × 200 mm) template, use the Lasercut Jigsaw extension to generate puzzle pieces, save as SVG, and observe/produce laser-cut versions. Optional post-cut coloring with acrylic paint pens. The day emphasizes the engineering design process while embedding Arizona 6th grade math standards directly into design decisions and a dedicated worksheet. Students experience how vector technology and laser manufacturing turn mathematical models into physical products.

Learning Objectives By the end of the field trip, students will be able to:

- Create and refine a vector-based puzzle design in Inkscape that incorporates artistic elements and is mathematically scaled for laser cutting.
- Apply 6th grade geometry and ratio skills (area calculations, coordinate plotting, proportional scaling) to solve real-world design problems.
- Follow the engineering design process to develop, test parameters, and optimize a solution for a functional puzzle.
- Explain how science/engineering knowledge (models, precise measurements) enables technology like laser cutting to create

products.

- Reflect on connections to careers in graphic design, manufacturing, and STEAM fields.

Arizona Standards Addressed

Mathematics (explicitly implemented via worksheet and design tasks):

- 6.G.A.1: Find the area of right triangles, other triangles, special quadrilaterals, and polygons by composing them into rectangles or decomposing them into triangles/other shapes; apply to real-world context (e.g., calculating areas of puzzle shapes or the overall design for material efficiency and fit within the 15 cm × 20 cm template).
- 6.G.A.3: Draw polygons in the coordinate plane given coordinates for the vertices; use coordinates to find side lengths (vertical/horizontal joins); apply to real-world context (e.g., plotting and verifying puzzle elements for precise laser paths).
- 6.RP.A.3: Use ratio and rate reasoning to solve problems in real-world context (e.g., scaling artwork proportionally to exactly fit 150 mm × 200 mm using tables of equivalent ratios, proportions, equations, or double number lines).

Science & Engineering (integrated via practices and core ideas):

- Science and Engineering Practices (SEPs): Develop and use models (digital vector designs and coordinate plotting); Use mathematics and computational thinking (scaling, area, coordinates); Construct explanations and design solutions (optimizing puzzle parameters in Inkscape and laser settings); Plan and carry out investigations (testing jigsaw extension variables).
- Core Idea U2: The knowledge produced by science is used in engineering and technologies to solve problems and/or create products (students see how precise math models enable laser technology to produce a tangible puzzle).
- Engineering Design Process : Ask/define problem →

Imagine/brainstorm → Plan (math + sketch) → Create (Inkscape + extension) → Test/optimize (parameters, scaling) → Improve/reflect.

These standards are actively practiced during the field trip through guided worksheet sections, software application, and laser observation.

Materials (per student or small group)

- Puzzle Design Worksheet (PDF – printed or digital; includes sections for area calculations, coordinate plotting, scaling ratios, and reflections).
- Computers/laptops with Inkscape (free vector software) and Lasercut Jigsaw extension pre-installed.
- Projector or shared screen for quick video demos (drawing shapes ~8 min, Lasercut Jigsaw ~7 min, laser/SVG overview ~2 min).
- Art inspiration images/slideshow (Navajo rugs, pottery, pictographs, modern geometric patterns).
- 15 cm × 20 cm template reference (physical or on-screen).
- Acrylic paint pens (optional, for post-cut coloring).
- Student sketches or brainstorming paper.
- Safety gear for laser area (as per site protocols – goggles, supervision).
- Storage for SVG files and finished puzzles will be done on computers via Moodle assignment.

Detailed Schedule (4-5 Hours Total)

0-15 min: Arrival, Safety Briefing & Engagement (Whole Group)

- Welcome and site safety rules (especially laser operation – never operate without adult supervision).
- Student's will take Career Inventory to assess their interests.

- Quick hook: Show examples of laser-cut puzzles and discuss real-world applications (custom toys, educational tools, art). Introduce vector graphics (sharp, scalable via math) vs. raster images.
- Overview of the day and engineering design process. Preview art inspirations and how math makes the design “laser-ready.”

15–75 min: Math Preparation with Worksheet (60 min – Core Math Implementation)

- Students work individually or in pairs on the Puzzle Design Worksheet .
- Area section (6.G.A.1): Decompose/compose sample puzzle shapes or their own sketched ideas into rectangles/triangles; calculate areas and relate to the 15 cm × 20 cm template (real-world: material use and fit).
- Coordinate plane section (6.G.A.3): Plot polygon vertices for puzzle elements; calculate side lengths using same x- or y-coordinates; discuss precision for laser cutting.
- Scaling section (6.RP.A.3): Use ratio reasoning, tables, or proportions to scale a design idea to exactly 150 mm × 200 mm. Practice with equivalent ratios or equations.
- Teacher circulates for support and formative checks. This block grounds the entire design in Arizona math standards before digital work begins.

75–195 min: Technology & Design in Inkscape (2 hours – with short break if needed)

- 15–30 min Demo/Review: Quick video clips or live demo – drawing shapes (circles, rectangles, triangles, stars, hearts), adding text/patterns, coloring, and manipulating objects. Show art examples for inspiration.
- 60–75 min Guided Creation: Students brainstorm themes, sketch quickly if needed, then build designs in Inkscape. Incorporate geometric patterns inspired by cultural/modern art. Use math from

worksheet (coordinates for placement, area awareness for balance).

- 30–45 min Jigsaw & Scaling: Apply Extensions → Render → Lasercut Jigsaw. Set width/height to 150 mm × 200 mm, adjust corner radius and number of pieces (across/down). Re-check scaling with ratios if adjustments are needed. Test parameters to optimize (engineering iteration).
- Save as Plain SVG (*.svg) with student name.
- Teacher support for troubleshooting; rotate groups if computer access is limited.

195–255/300 min: Laser Cutting Production, Coloring & Reflection (60–105 min)

- Group Demo/Observation: Teacher or site expert demonstrates the laser cutter using student SVGs (or sample files). Explain how the machine reads the vector paths and math-based design for precise cuts/engraving. Discuss real-world manufacturing (U2 connection).
- Hands-on Production: Process student files in small batches (depending on machine capacity). Students observe or assist with safe loading/unloading under supervision.
- Optional Coloring: Use acrylic paint pens to decorate cut pieces (quick-drying, no cleanup).
- Reflection (15–20 min): Whole group or written – How did ratios and coordinates help your design work with the laser? What engineering choices did you make? What careers might use these skills? Share one art inspiration and one math connection.

Total Timing Notes:

- Flexible buffer for transitions, questions, or extra laser time (aim for 4 hours minimum; extend to 5 with deeper reflection or multiple cuts).
- Include a short lunch/snack break if the schedule spans mealtime.

- If computer time is constrained, prioritize worksheet math first, then focus Inkscape on key elements (shapes + jigsaw) rather than highly detailed art.
- Important: If time is an issue, instructor will pre-cut puzzles that students will then engrave their designs on.

Assessment

- Formative: Worksheet completion and accuracy (rubric sections for 6.G.A.1 area work, 6.G.A.3 plotting/calculations, 6.RP.A.3 scaling – with real-world explanations). Observation during Inkscape (evidence of iteration and math application).
- Summative: Completed SVG file (technical quality, proper scaling/dimensions, jigsaw application) and optional physical puzzle. Reflection response linking math, engineering, and technology.
- Success Criteria: Design fits 15 cm × 20 cm template mathematically, is laser-ready, incorporates creative elements, and students can articulate standards connections.

Differentiation and Accommodations

- Support: Pre-drawn templates or simplified shapes; paired work; printed worksheet with visuals; teacher-guided stations. Pre-cut Puzzle pieces.
- Extension: More complex multi-layer designs, additional pieces, or calculate material waste using areas. Advanced students help peers or explore Inkscape patterns.
- Accessibility: Visual demos, video captions, vocabulary support (vector, decompose, ratio, coordinate, SVG), extended time for worksheet or design.
- Field Trip Considerations: Account for travel time, group sizes for laser access, and behavioral expectations at the host site.

Extensions & Cross-Curricular Ties

- Discuss environmental impacts of manufacturing or cultural respect in art inspirations.
- Career connection slideshow/questions

Safety Notes

- Strict adult supervision for laser cutter (eye protection, no direct operation by students).
- Standard computer lab and field trip safety protocols.
- Emphasize respectful use of cultural art inspirations.

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